Scratch: Variables, Score & Lives

Give your game a memory: scores, lives and speed!

Scratch | Beginner | Ages 6–12

InfiBee Coders

💡 The Big Idea

- A variable is a named box that stores a value, such as a number or some text.
- The value can change while your game runs, so the box is a great place for a score.
- Games use variables for score, lives, level, speed and timers.
- You make one in the orange Variables category with Make a Variable.
- Always give a variable a clear name like `score` or `lives`.

🧰 Making and Using Variables

- Click Make a Variable, type a name and choose For all sprites or For this sprite only.
- `set [score] to (0)` puts an exact starting value in the box.
- `change [score] by (1)` adds to the box. A negative number takes away.
- Tick the box beside a variable to show it on the stage, or use `show variable` and `hide variable`.
- Drag the round variable block into other blocks, like `say (score)`.

🎮 Score, Lives and Game Over

- Set the starting values when the green flag is clicked.
- Add to `score` when the player catches or collects something.
- Take 1 from `lives` when the player touches danger.
- Use `if <(lives) = (0)> then` to decide when the game is over.
- Use `stop [all]` to end the game once it is over.

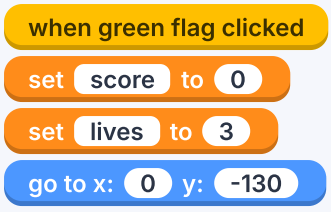
🌍 Variables in Real Life

- A scoreboard at a cricket match is a variable that goes up with every run.
- A piggy bank total changes each time you add or take out money.
- The battery percentage on a phone is a variable that goes down as you use it.
- A video game health bar is a variable that shrinks when you get hit.

💻 Script Examples

Start of a game

Scratch blocks

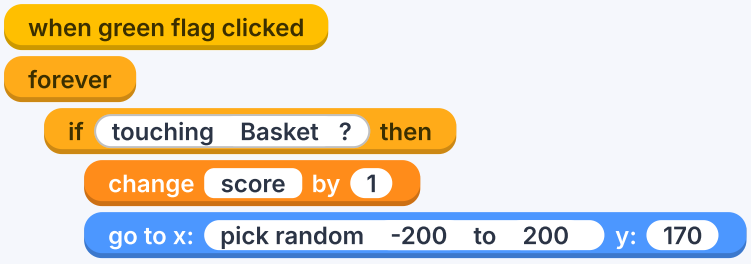


▶ What happens:

The score shows 0 and lives shows 3 every time the game starts.

Catch a falling apple (on the Apple sprite)

Scratch blocks

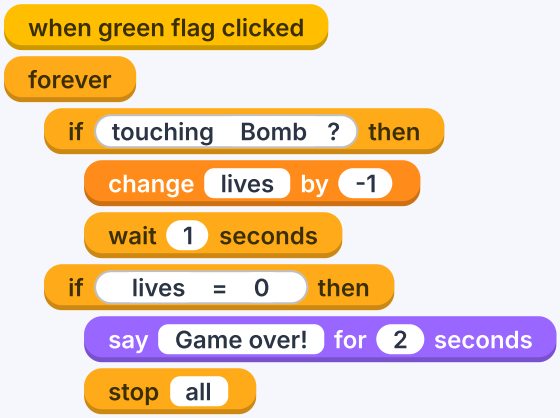


▶ What happens:

Each catch adds 1 to the score, and the apple jumps to a new spot at the top of the stage.

Lives and game over

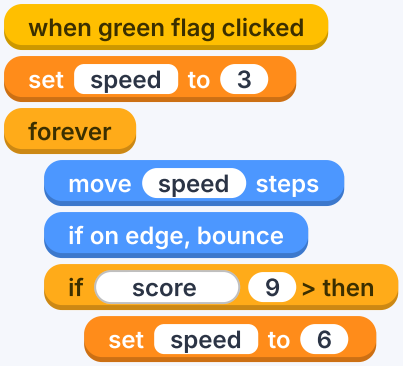
Scratch blocks



▶ What happens:

A bomb costs one life, and the one second wait stops it taking all the lives at once. At 0 lives the game ends.

Speed up as the score grows — Scratch blocks



▶ What happens:

The sprite moves at speed 3, then doubles to 6 once the score goes above 9.

Show the score in a speech bubble — Scratch blocks



▶ What happens:

If score is 5, the sprite says: Score: 5

📚 Key Terms

Variable
A named box that stores a value that can change.

Value
The number or text inside a variable.

Set
Put an exact value into a variable.

Change
Add to (or take away from) the value in a variable.

Score
A variable that counts points.

Lives
A variable that counts how many chances are left.

For all sprites
A variable every sprite can read and change.

For this sprite only
A variable that belongs to one sprite.

Reporter
A rounded block that holds a value, like a variable.

Condition
A true or false test used inside if blocks.

🧮 Quick Reference

`set [score] to (0)`	start from a value
`change [score] by (1)`	add 1
`change [lives] by (-1)`	take away 1
`<(lives) = (0)>`	is it game over?

```
<(score) > (9)>              is the
score above 9?
```

```
pick random (1) to (10)      a surprise
number
```

```
say (join [Score: ] (score)) show text
and a number
```

✏️ Worked Examples

Q: Make the score go up by 5 each time a coin is touched.

A: Step 1: Make a variable called score. Step 2: Under the green flag add set [score] to (0). Step 3: Inside forever, add if <touching [Player]?> then. Step 4: Put change [score] by (5) inside the if.

Q: Give the player 3 lives and end the game at 0.

A: Step 1: Make a variable called lives. Step 2: set [lives] to (3) when the flag is clicked. Step 3: change [lives] by (-1) when something bad touches the player. Step 4: if <(lives) = (0)> then stop [all].

Q: Show Score: 12 in a speech bubble.

A: Use say (join [Score:] (score)). The join block glues the words and the number together, so a score of 12 gives Score: 12.

🚫 Common Mistakes

- Forgetting to set the score to 0 at the start, so it keeps the number from the last game.
- Putting `set [score] to (0)` inside the `forever` loop, which resets the score every moment.
- Using `change` when you meant `set`, or the other way round.
- Making a variable For this sprite only when other sprites need to use it.
- Choosing unclear names like `x` or `thing` that you forget later.

👨‍👩‍👧 Parent Corner

- Ask: What does the score start at? What makes it go up? What makes it go down?
- Play the game together, then ask your child to change one number, like 3 lives to 5, and see what happens.
- Ask them to add a second variable, like a high score or a timer.
- Connect it to maths: taking away a life is adding a negative number.
- Celebrate the logic they used, not only the finished game.

🎯 Quick Check

1. What does set [score] to (0) do?

✅ It puts exactly 0 in the score variable, a starting value.

2. Which block adds 1 to a variable?

✅ change [score] by (1).

3. How do you take away one life?

✅ Use change [lives] by (-1).

4. Why put set [score] to (0) under the green flag?

✅ So every new game starts fresh at 0.

5. Where do you find variable blocks?

✅ In the orange Variables category.